DON BOSCO INSTITUTE OF TECHNOLOGY BENGALURU

Department of CSE(AI&ML) Groups No.:17

Academic year:2024-25 (EVEN)

Major Project (BCI685) (6th Semester)

Project Title: Smart Agriculture System with Machine Learning

Abstract:
Agriculture is the backbone of many economies, and technological advancements have led to the evolution of precision farming. This project focuses on developing a **Smart Agriculture System using Machine Learning**, without the use of IoT devices. The system leverages historical and real-time data, including weather conditions, soil properties, and crop growth patterns, to optimize farming practices. By applying machine learning algorithms, it provides predictive insights into crop yield, disease detection, soil health assessment, and crop recommendation. This system aims to assist farmers in making data-driven decisions, reducing losses, and maximizing agricultural productivity.

Keywords: Smart Agriculture, Machine Learning in Agriculture, Plant Disease Detection

Problem Statement:

Traditional farming methods rely heavily on experience and intuition, leading to inefficiencies and unpredictable outcomes. Farmers often struggle with:

- Inaccurate Crop Selection – Choosing unsuitable crops for their soil and climate conditions.
- Uncertain Yield Estimation – Lack of data-driven predictions on potential harvest output.
- Late Disease Detection – Inability to detect plant diseases early, causing yield loss.

Objectives:

1. **Crop Yield Prediction** – Estimate potential yield based on soil properties, weather, and historical data.
2. **Crop Recommendation System** – Suggest the most suitable crops for cultivation based on environmental conditions.
3. **Disease Detection** – Identify plant diseases using image-based machine learning techniques.

Outcome:

- Improved Crop Selection – AI-driven recommendations based on soil and climate conditions.
- Accurate Yield Predictions – Data-driven insights for better planning and resource allocation.
- Early Disease Detection – Image-based models to identify plant diseases and suggest remedies.

Language / Software Tools used:

Frontend(HTML , CSS , Javascript), Backend(Python and Libraries), VS code

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